Spellbrawl: what to change and improve

13 October 2026 · Reflects the Spellbrawl code on that date, including the downtime and lag-compensation changes made that day

Spellbrawl is the knockback-only mode added on 12 October. Between 2 and 12 mages each take three spells from a pool of nine, plus the shared Ember Bolt, and shove each other off a crumbling basalt island into lava. Spells don't hurt. Lava burns 14 health per second, and the lower your health, the further you fly. This document collects what to change and build to make that mode the whole game for a Steam release, and names the code each change touches.

1 The plan at a glance

1. Shorter downtime between rounds, and a lag-compensation cap for hits on players. DONE §2
2. An island renderer built from Voronoi cells for the Caldera, then the Steam store page, so wishlists start building. §8, §9
3. Server-side bots. §7
4. Momentum transfer between flung players, deflecting projectiles, and hit feel. §4, §9
5. Spell augments in place of flat boons, and about eight new spells. §5
6. The Rising Tide map, then team modes. §6
7. Steamworks networking, controllers and local play, then a Steam Playtest and a Next Fest demo with bots. §9

2 Where things stand

Shorter downtime DONE

Phase	Before	Now
Draft	40 s	20 s, and the clock waits while only one player is present
Countdown	3 s	3 s
Round result	4.5 s	3.5 s
Shop between rounds	25 s	12 s
Most time outside fights in a 5-round match	177.5 s	100.5 s

When everyone presses Ready, the next phase still starts at once. A host waiting for friends is never pushed into a solo round, and Ready still starts a round alone. The draft and the shop already doubled as practice: shoves work in both phases, and lava only burns during a fight. The constants live in `Brawl.h`; the waiting clock is in `MatchDirector::UpdateBrawl`.

Lag compensation capped for hits on players DONE

Hit tests now look back at most 8 ticks (133 ms) for where a player target was: `MaxPlayerRewindTicks` in `Components.h`, applied in `TargetPosition` in `Combat.cpp`. Monsters keep the 28-tick limit. At 150 ms round-trip ping the server looks back 11.35 ticks on average, so players under about 95 ms get full compensation. At 150 ms a shooter has to lead a running target by about 0.3 units more, which is less than a player's radius. A high-ping shooter's client can still draw an impact that the server rejects.

Tests: the brawl harness gained downtime and player-rewind suites, and all of it passes. The PvE lag-compensation test still lands 35 of 35 arrows.

Camera DECIDED

The player-centred camera stays. If hits from off-screen become a complaint, edge indicators for rivals and incoming spells are the light fix.

3 Why build the whole game around Spellbrawl

Scope fits one developer. A loot ARPG has to keep producing items, enemies, maps and an endgame. A round-based brawler mostly needs spells, maps and modifiers, and players replay each one hundreds of times.

The formula is proven. It's the Warcraft III custom map *Warlock* plus the catch-up rule from *ROUNDS*, where the round winner gets no power-up. Spellbrawl already works that way. Knock-them-off party games such as *Stick Fight*, *Gang Beasts* and *Bopl Battle* sell on Steam because friend groups and streamers carry them.

The tech fits. The arena is small. The collapse is computed from `CollapsePlan` and the round clock on both server and client, so it costs no bandwidth. Lava as the main light source shows off the radiance cascades better than any other setting.

The real risk is player count. Warlock-likes that depended on matchmaking, such as *Spellsworn* and *Magicka: Wizard Wars*, faded once matchmaking ran dry. Design it so a friend group never needs strangers: bots, Steam invites, and local multiplayer, which also enables Remote Play Together.

Reuse the ARPG. Emberhold becomes the lobby, with practice dummies. The monster AI and telegraphs become a co-op mode where you push monsters into the lava. The KayKit gear becomes cosmetics.

4 Skill expression

Keep the floor low for a party crowd, and build the ceiling from mechanics rather than extra buttons.

4.1 Bodies as weapons

Players pass through each other today. In Brawl, a flung player whose push is above `PushStagger` (12) and who touches another player should pass on about 60% of the push and keep the rest. The knockout goes to whoever cast the original spell. That makes bowling-style multi-knockouts, the best clip moments this genre has.

It is a server-only check after player movement: flung players against player bodies, with a short per-pair cooldown so one contact transfers once. The second victim simply receives a push, like any spell hit, so prediction needs no change.

4.2 Deflect

Gust bounces projectiles in its cone back toward the caster's aim, Shockwave bounces them outward, and two projectiles that collide cancel, with the heavier one surviving. That adds a timing skill to spells players already own. Mirror Ward's `Reflect` path already reverses a projectile, hands it to its new owner and clears its hit list, so deflect can reuse it.

4.3 Steering while flung

`PushControl` already leaves 15% of movement control at full push (`PushControlLoss` is 0.85). Make that readable: holding against the push shortens it by up to about 20%, holding sideways bends it, and a trail behind the flung player shows it working. It runs in `SimulatePlayer`, so it is predicted like the rest of movement and costs nothing on the network.

4.4 Edge-guarding and reads

Warp is the recovery, and swapping out of the lava with Swap is already a clutch move. Show Warp and Mirror Ward readiness on the character, for example a rune that dims while the spell recharges, so "punish them after they've used Warp" becomes a read. The terrain spells in §5 let players deny a recovery spot.

5 Spells, augments and mutators

5.1 New spells

The pool of nine covers poke, burst, mobility, control and defence. Terrain and trick spells take it to 16–20 at launch. Most need new behaviour code, as Swap and Mirror Ward did, rather than just table entries.

Spell	Kind	What it does	What it needs
Fissure	Terrain	A line that cracks the cells it crosses; they collapse 1.2 s later	Per-cell collapse overrides (§6)
Obsidian	Terrain	Cools a patch of lava into crust for 4 s: a bridge or a self-rescue	Temporary cell overrides
Basalt Pillar	Terrain	Raises a pillar that blocks shots and stops knockback	A dynamic collider
Chain Hook	Trick	Pulls a rival to you, or you to them	New projectile behaviour
Boomerang	Trick	Shoves on the way out, pulls on the way back	A returning projectile path
Homing Wisp	Trick	A slow orb that tracks its target; lure it into someone else	Projectile steering
Geyser	Trick	A delayed eruption at a spot: a big launch plus burn	Mostly a ground ability (`PendingStrike`)
Recall	Trick	Returns you to where you were 1.5 s ago	A few past positions in the predicted player state
Rush	Trick	A dash that body-checks whoever it hits	A dash with a contact push

5.2 Augments instead of flat boons

The nine generic boons (+7% move speed, shoves 12% harder and so on) are the weakest part of the round loop. Replace most of them with spell augments in the style of *Hades* and *ROUNDS*, offered only for spells you own. Add a few risk cards, such as "your spells shove 40% harder, and you get shoved 30% further".

Spell	Example augment
Ember Bolt	Splits into three at the end of its range
Gust	Bounces projectiles in its cone back at their caster (the deflect from §4.2)
Warp	Leaves a shockwave where you left
Comet	The crater breaks the cells it lands on
Swap	Also swaps health, and with it the fling scaling
Gravity Well	Implodes when it ends, flinging everyone outward

`BoonDef` gains the spell it modifies, `OfferBoons` filters by the loadout, and the effects are read from the loadout the way `GetKnockbackScale` reads Heavy Hand and Anchored today. Keep the rule that the round winner gets no boon.

5.3 Mutators

Mutators are the cheapest content you can make: a few rules in `MatchRules`, which already travels with the session.

Mutator	Change
Ice	Much lower `PushFriction`, so shoves slide far
Hurricane	Every push ×1.5
Meteor Shower	Telegraphed comets land at random spots
Darkness	Lava is the only light, which shows off the radiance cascades
Random Spells	A new random loadout every round
Event Round	A random mutator every few rounds

6 Maps and modes

> **Keep world state a function of seed and clock**
>
> Today the collapse comes from `CollapsePlan` and `BrawlStatus.Clock`, computed identically on server and client, so it costs no bandwidth and both sides always agree. Rising lava, drifting rafts and scheduled eruptions fit the same model. Only terrain changes made by players (Fissure, Obsidian) need replicating, as a short list of (cell, time) overrides attached to the plan.

Map or mode	Idea	What it needs
Caldera (today)	The shrinking island, re-skinned as a real island (§8)	Renderer work only

Map or mode	Idea	What it needs
Rising Tide	A terraced island where the lava rises in steps; a cell counts as lava once the lava is above its top	One-way ledges in Brawl (`MoveCircle` blocks ledges both ways today); spells that can hit up onto terraces (`ProjectileHeight` blocks that today); the lava level as a function of the clock
Archipelago	Islets joined by bridges that collapse first, so mobility becomes the build	A map and a collapse plan that takes bridges first
Drift	Basalt rafts carried by lava currents	Raft positions from the clock; `SimulatePlayer` adds the velocity of the raft under you
Teams (2v2, 3v3)	A Chain Hook or Gravity Well on an ally pulls them out of the lava	Team rules from Arena; pulls that can target allies
Lava Ball	Knock a heavy orb into the other team's pit	A ball entity that takes pushes
Co-op Siege	Monster waves climb onto the island, and you push them in	Enemy AI, telegraphs and bosses from the ARPG; knockback on monsters

Teams are probably the competitive format. Free-for-all stays the party default.

7 Bots

7.1 Where they run

Run each bot inside the server tick as an ordinary `PlayerController` whose `PlayerInput` comes from a `BotBrain` instead of the network. They follow exactly the same rules as humans and need no loopback `GameClient` or prediction. Filling empty lobby slots with bots then becomes a lobby option. The bot in the test harness (random strafing, random casts) is only a smoke test.

7.2 Knockback has a closed form

Push speed decays exponentially, with k = `PushFriction` = 3.2 per second, so a shove travels a fixed distance:

$$v(t) = v_0\, e^{-kt} \;\Rightarrow\; d = v_0 / k$$

- Gust: 21 / 3.2 ≈ 6.6 units, or about 10.5 at 0 health (fling scaling up to ×1.6).
- Ember Bolt: 11.5 / 3.2 ≈ 3.6 units. The harness measures 3.61.

So for every spell and every rival, a bot can compute where a hit would send them, and where it would land itself if hit. Attacking and positioning use the same function.

7.3 How a bot decides

About every 100 ms:

- **Safety:** score spots on the island by lava, cells already showing their collapse warning (only what a human could see), projectile paths and ground telegraphs.
- **Positioning:** keep the centre of the island behind you and rivals between you and the edge.
- **Casting:** aim at the intercept point, and cast when the chance of hitting times the value of the push is high enough.
- **Defence:** Mirror Ward against a projectile landing within about 0.2 s, otherwise roll sideways.
- **Recovery:** in the lava, Warp to the nearest cell that isn't about to collapse.

7.4 Difficulty and personality

Difficulty comes from reaction delay, aim noise, how far ahead the bot reads the collapse, and whether it uses combos like Gravity Well into Comet. Personalities are just different weights: aggressive, sniper, trickster. The draft and the shop use archetype loadouts and simple boon picks.

7.5 The real payoff: balance runs

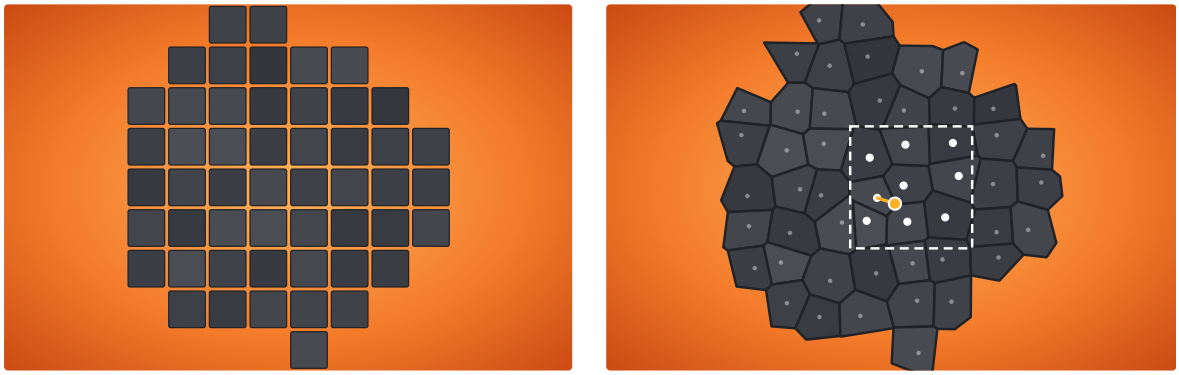
Headless bot-against-bot runs overnight can report win rates per spell, round lengths and how players die. With 20 spells and their augments, balance can't be done by hand. The harness already plays whole matches headless with 2, 4 and 8 players; the bot brain is the missing piece.

8 Visuals: an island instead of cubes

Keep the grid as the data structure, and change the shape of each cell.

8.1 Voronoi cells

Give each tile one seed point, jittered by up to ±0.3 of a tile around its centre, and let the tile become the Voronoi cell around that seed. Finding the cell under a point is a nearest-seed search over the 3 × 3 surrounding tiles, and with jitter of at most ±0.3 that search is exact. `TileCollapseTime`, indexing and networking stay as they are. The rendered pieces are exactly the gameplay pieces, so where you see rock is exactly where you don't burn.



Today: one square slab per tile

Proposed: one jittered Voronoi cell per tile

Figure 1. The same 14 × 9 tile grid and the same island tiles. Left, each tile is a square slab, as today. Right, each tile gets one jittered seed (the dots) and becomes the Voronoi cell around it. The dashed square is the 3 × 3 block searched for the cell under the orange point, and the line goes to the nearest seed, which here belongs to a neighbouring tile.

```
// CellSeed puts each seed in [0.2, 0.8] of its tile, so the 3 x 3 search is exact.
TileCoord CellAt(Vec2 p, uint32_t seed)
{
    const int32_t tx = int32_t(std::floor(p.x));
    const int32_t ty = int32_t(std::floor(p.y));
    TileCoord best{ tx, ty };
    float nearest = FLT_MAX;
    for (int32_t dy = -1; dy <= 1; dy++)
        for (int32_t dx = -1; dx <= 1; dx++)
            if (const float d = LengthSquared(p - CellSeed(tx + dx, ty + dy, seed)); d < nearest)
            {
                nearest = d;
                best = { tx + dx, ty + dy };
            }
    return best;
}
```

A hex layout of seeds instead gives columnar basalt, as at the Giant's Causeway. That suits Rising Tide, where cells sit at different heights.

8.2 One island mesh

Build the island mesh at load. Each cell is a prism. Its top vertices come from a smooth noise heightfield, so neighbouring cells share corners and the surface reads as one flat-shaded, low-poly landmass. Its sides extend below the lava, with a noisy bottom ring for cliffs. Colour by height and by distance from the edge: ash or dry grass inland, then basalt, then obsidian with a glowing heat band at the lava line.

Store a cell index on each vertex and keep a small per-cell state buffer: drop, tilt axis and angle, heat, warning. The whole island is then one draw call that crumbles in the vertex shader. A few hundred cells at about 30 triangles each is nothing for `WorldRenderer`.

8.3 Cracks, chunks and the shoreline

- **Warning cracks:** draw them in the island shader from the distance to the cell edge, instead of per-tile Fracture decals, so the warning traces the exact outline of what will fall.
- **Chunks:** group neighbouring cells with similar collapse times so they tilt over their outer edge and sink together. Add a lava splash, embers, a glow pulse that the radiance cascades pick up, and ripples in the lava shader from a short list of ring sources. It reads as the island breaking apart rather than tiles vanishing.
- **Shoreline:** a small texture of island height or distance to the edge lets the lava pass (`Lava.frag`) draw a glowing crust band where lava meets rock. The same texture drives the Rising Tide shoreline.

8.4 Polish

- Props ride their cell down into the lava.
- The reset between rounds becomes a moment, with fresh rock rising out of the lava.
- Big flings show as a visual arc, using the `Flung` flag that is already replicated.

9 Steam release essentials

- **Steamworks lobbies, invites and Steam Datagram Relay.** The game uses GameNetworkingSockets, which has the same API as Steamworks' `ISteamNetworkingSockets`, so switching is small. The relay removes port forwarding and hides players' IP addresses; hosting is direct IPv4 today. In a party game, joining a friend has to take one click.
- **Local multiplayer and controllers.** Several players on one machine enable Remote Play Together, where only the host owns the game and friends join over a stream. With the player-centred camera, couch play needs either a shared camera used only when several players share a screen, framing all of them and zooming out as they spread, or split screen.
- **Juice for clips:** a brief freeze on big hits, a slow-motion zoom on each round's final knockout, and a distinct colour per player on their spells.
- **Store page and tests:** put the store page up once the island looks good, so wishlists start building. Then run a Steam Playtest and show a Next Fest demo with bots.

10 Reference numbers

Current values, for planning and tuning.

What	Value
Simulation	60 Hz, up to 12 players, protocol v12
Health	100, plus 20 per Vigor boon
Lava	14 health per second, then 0.8 s of afterburn at 8 per second
Fling scaling	× (1 + 0.6 × missing health fraction), up to ×1.6
Push	Decay `PushFriction` 3.2 per second, cap `MaxPush` 42, interrupts casts above `PushStagger` 12
Movement	5 units per second; dodge every 2.4 s
Island radius	6.5 + 1.1 per player, clamped to 8.5–12
Collapse	7 s grace, then the edge recedes 0.16 units per second; the final collapse starts at 78 s
Round length (bot matches)	27–68 s
Content	9 spells plus Ember Bolt; 9 boons
Time outside fights	At most 100.5 s per 5-round match (was 177.5 s)
Lag compensation	8 ticks against players, 28 against monsters